

MLFF Architecture Revision 85

CUEQ-REPEAT1-DIAG3 warm-up + all-pairs repeatability refinement

mdstats project

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Revision 85

Release: mdstats 0.20.218a0

Dependency-graph schema: 67

Gate: CUEQ-REPEAT1-DIAG3

The revision-84 MPA-0/default workstation evidence confirmed that TRAIN2 FP32 force differences are dominated by a stochastic execution envelope rather than a stable e3nn/pure-CuEq offset. In ordinary execution, e3nn-self **Fmax** reached $1.962\text{e-}5$, CuEq-self $1.836\text{e-}5$, and cross-backend $2.270\text{e-}5$. Under the isolated deterministic controls, e3nn-self comparisons against run 1 became a constant $1.061\text{e-}5$, while CuEq-self still reached $2.709\text{e-}5$ and cross-backend $1.550\text{e-}5$. Selection remained identical throughout.

The constant deterministic e3nn self value exposed a second problem with the temporary diagnostic: every self comparison used run 1 as its reference, so a first-call/warm-up shift could contaminate all nine reported self comparisons. Revision 85 removes that arbitrary baseline.

Each backend now receives one complete **discarded warm-up evaluation** on the exact doctor probe. Ten post-warm-up outputs are then retained for e3nn and ten for pure CuEq. No additional model calls are needed for pair construction. mdstats computes offline:

- $C(10,2) = 45$ e3nn-self pairs;
- $C(10,2) = 45$ CuEq-self pairs;
- $10 \times 10 = 100$ e3nn/CuEq cross-backend pairs.

Every pair records **Fmax**, force RMSE, force absolute-error p99 and p99.9, the number of force components above $1\text{e-}5$, energy/stress/descriptor maximum errors, and selection identity where applicable. Terminal summaries report **min**, **median**, **p90**, **p99**, and **max** instead of printing all 100 cross pairs.

The ordinary diagnostic record advances to **mdstats.training-acceleration-repeatability-diagnostic.v2**. Historical v1 records remain readable and are interpreted using their original baseline-comparison semantics. The isolated deterministic-control subprocess remains schema v1 at the outer wrapper but carries the new v2 repeatability record when completed.

Revision 85 remains **non-authorizing**. TRAIN2 FP32 parity stays at **rtol=1e-5**, **atol=1e-5**; source/DATA6 and FP64 tolerances are unchanged. FINAL-GPU1 remains archival until the DIAG3 workstation distributions are reviewed and a permanent noise-normalized backend-equivalence criterion is frozen.